Yeast Fermentation and Fuel Sources

Teacher guide | Biology 2e, Chapter 7 | 45-60 minutes

Learning targets and timing

How does the available sugar affect gas accumulation in a yeast culture?

Prepare fresh mixtures with no added sugar, glucose, and sucrose. Connect each flask to a gas syringe, observe for 30 simulated minutes, and collect three trials per treatment. Then explore temperature or a no-yeast control separately. Compare average gas accumulation rates over the same time interval.

Suggested sequence: 8-10 min prediction and setup; 20-30 min observation; 12-20 min analysis. Virtual time can be accelerated. For photosynthesis and yeast, pool independent replicates from class groups; three sequential repeats need extra class time.

Route choices and sharing

The student pages open as guided virtual investigations. Teacher customization is deferred to a future teacher workspace. Use this worksheet to assign classroom components separately; discuss data-source differences when comparing results. Existing hybrid links still work, but the student page no longer creates them.

Materials

- Baker's yeast, glucose, sucrose, water, balance, thermometer and timer

- Three matched flasks and tubing connected to freely moving 100 mL gas syringes

- Teacher-managed 30 deg C water bath, separate yeast and sugar stocks, plain-water control; use the classroom handout for stock and flask volumes

Classroom procedure

1. Revised classroom target: 20 g yeast and 80 g sugar per 1,000 mL total water. Prepare separate stocks: 20 g yeast in 500 mL water, and 80 g glucose OR sucrose in another 500 mL water. For larger stocks use 40 g yeast in 1,000 mL water and 160 g of each sugar in its own 1,000 mL water.

2. These are water-added recipes, not final solution volumes. Equal measured stock aliquots are an approximation; equal fractions of matched stock batches preserve the exact per-water recipe. Keep the no-added-sugar control and all stock labels distinct.

3. Prepare sugar stocks ahead; hydrate yeast close to use following package guidance and gently remix before every aliquot. Add yeast last. Suggested equal-aliquot fills: 10 mL treatment stock plus 10 mL yeast stock in a 50 mL flask, or 20 + 20 mL in a 100 mL flask. Keep temperature and method consistent.

4. Use the selected classroom collector and its handout. For balloons, record circumference at 0, 10, 20, 30, 40, 50 and 60 minutes; record NI when not inflated. Volume collectors use 0-30 minutes at 5-minute intervals, stopping before capacity. Extend only under a shared teacher-selected schedule. The stronger recipe does not guarantee a shorter inflation delay.

5. Pool three fresh independent mixtures per treatment. Keep balloon centimeters, collected-gas milliliters, and the lower-concentration virtual results separate. Use the downloadable classroom sheets for longer observations; the online virtual-model grid remains 0-30 minutes.

Safety: Use baker's yeast only. Do not taste cultures. Never seal gas production in a rigid closed container or block the syringe plunger. Stop before apparatus capacity is reached. Teacher checks glassware and warm-water handling.

Yeast Fermentation and Fuel Sources

Teacher guide / interpretation | Biology 2e, Chapter 7 | 45-60 minutes

Expected reasoning and troubleshooting

Virtual expectation: both sugar treatments accumulate more gas than the no-added-sugar control. Glucose and sucrose may exchange rank across trials. Encourage evidence-based comparison rather than a fixed "winning sugar."

Fermentation regenerates NAD+ so glycolysis can continue. Fermentation and anaerobic respiration are different pathways; yeast gas volume does not uniquely establish which metabolic processes occurred.

No-added-sugar is not necessarily zero because yeast has internal reserves and apparatus has background effects. A no-yeast control and leak/temperature checks strengthen the inference.

Model boundaries

Volumes are illustrative, with a lag, gradual startup, and trial variation. Revised kinetics illustrate substrate availability; the minute-scale parameters are not fitted to experimental data. Saved older investigations retain their earlier kinetics. The virtual water bath changes gradually; temperature effects from 15 to 40 deg C are simplified and omit thermal gas expansion, yeast damage, and substrate depletion. Glucose and sucrose responses overlap; their ranking is not universal. The water control can show small background gas. The model does not establish anaerobic conditions or identify collected gas chemically.

Average gas accumulation rate = (final volume - initial volume) / elapsed minutes. Gas from yeast is consistent with CO_2 production, but volume alone does not prove fermentation rather than aerobic metabolism or directly measure ATP yield.

Suggested assessment - 12 points

Dimension	Evidence to look for	Points
Design	Reasoned prediction; variables, controls or sampling plan	0-3
Data	Complete observations, correct units, source labels, repeats	0-3
Analysis	Appropriate calculations and comparisons supported by data	0-3
Explanation	Biological reasoning, uncertainty, and a feasible improvement	0-3

For each dimension: 0 = absent; 1 = partial or major errors; 2 = mostly sound with a minor omission; 3 = complete and well supported. Suggested teacher rubric only; the app does not assign a grade.

Downloads, access, and classroom pilot

Student worksheet: preparation, investigation, and exploration pages where included, plus companion cutouts. Teacher guide: procedures, interpretation, and rubric. Reports and CSV data download from Analyze & explain. Teacher guides are public resources, not protected answer keys. Before classroom use, pilot the equipment, timing, and reading level. Editable Word and fillable PDF versions are planned.

Reading and method connections

Scientific evidence and simulation assumptions: Open source

Sucrose physiology varies by strain (Rodrigues et al., 2021): Open source

Mechanism of sucrose utilisation (Mwesigye and Barford, 1996): Open source

OpenStax Biology 2e, 7.5: Open source

The investigation materials are original; appended cutouts were supplied by the classroom teacher. Simulated observations are not measurements from the linked sources.

Yeast Fermentation and Fuel Sources

Teacher guide / interactive gas collection | Biology 2e, Chapter 7 | 45-60 minutes

Preparation first; exploration after comparison

Students choose the assigned fuel, prepare the 50 mL mixture with 0.5 g yeast, and connect the stopper and tubing to the movable gas syringe. The flask stays empty until preparation is finished. Each treatment has three guided repeats at 30 C for 30 simulated minutes. No-added-sugar is preselected and needs no extra confirmation.

The timer runs at 30x, 60x, or 120x, with pause and time-advance controls. It stops at 30 minutes. Students must click Collect data and continue to save the trial and advance. Data are recorded every five minutes. Completed trials and classroom data are preserved on this device; reload restores an unfinished virtual trial paused.

Free exploration

After all nine trials, students can adjust the bath from 15 to 40 C. The simulated temperature approaches the setting gradually and affects subsequent gas accumulation. Students can resume past 30 minutes, up to 120 minutes or 100 mL. Use a fresh mixture to change fuel or omit yeast. Explore a no-yeast control and discuss physical gas expansion, which this simplified model does not represent.

Exploration is stored separately and exported with fuel, yeast presence, time, gas volume, and temperature. It never replaces the nine comparison trials. Download exploration before starting a fresh mixture; that action replaces the current exploration.

Interpretation and model limits

Assess predictions, matched conditions, measured rates, variability, and evidence-based explanations. The plunger shows cumulative gas volume, not foam height or ATP yield. This illustrative model has trial variation and a lag but omits substrate depletion, leaks, thermal expansion, and temperature damage. It does not prove oxygen was absent or establish that all gas arose through fermentation. Never block a real syringe or seal a rigid vessel against gas production.

Cut & sort: matter and energy

Optional introduction or extension | 10-15 minutes | Biology 2e, chapters 7 and 8.

Choose Card Activity on the lab overview to open the separate circular photosynthesis and respiration model, or arrange the printed cards on chart paper. Use two molecule sets for both halves at once, or reuse one 19-card set as you switch processes.

Print cutout pages in color, single-sided, at 100% scale. Cut along dashed lines. Use scissors safely. Group work: one person arranges, one counts atoms, and one explains; then rotate roles.

Model aerobic cellular respiration. Reuse the molecule cards from the photosynthesis sheets and add the three respiration symbols. Set aside any symbols not needed.

Count carbon, hydrogen, and oxygen atoms separately on each side. Explain how your arrangement shows conservation of matter and the transfer of energy. Energy symbols are annotations, not part of this simplified atom inventory.

This equation models aerobic respiration, not yeast fermentation. Fermentation does not use oxygen. Respiration begins in the cytoplasm; most later stages in plant and animal cells occur in mitochondria. Plants respire too.

Teacher use: assign this before prediction or after analysis. Accept either order of molecule groups on the same side. Ask students to explain placement, not just match a key. The online model checks each group and atom counts for practice feedback only; no grade is submitted.

The supplied originals follow, including answer keys. Student packets omit the photosynthesis teacher page and the answer text under the respiration cutouts. For respiration, the ATP symbol does not specify yield; ADP and phosphate are not shown.

PHOTOSYNTHESIS • CUT & SORT

Cut on dashed lines. Keep all three sheets together.

Build a balanced equation. Place light and chlorophyll where they help.

O_2



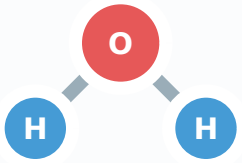
Oxygen

CO_2



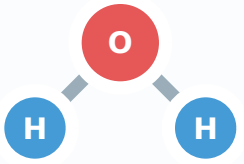
Carbon dioxide

H_2O



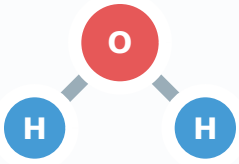
Water

H_2O



Water

H_2O



Water

O_2



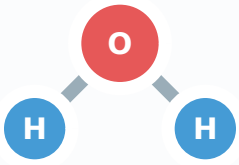
Oxygen

CO_2



Carbon dioxide

H_2O



Water

CO_2



Carbon dioxide

Atom colors: C = gray • O = red • H = blue

Student cutouts 1/3 • Print in color, single-sided, at 100%.

PHOTOSYNTHESIS • CUT & SORT

Cut on dashed lines. Keep all three sheets together.

Build a balanced equation. Place light and chlorophyll where they help.

+

Combine with

O_2



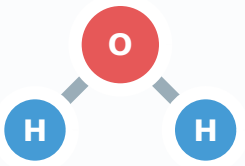
Oxygen

CO_2



Carbon dioxide

H_2O

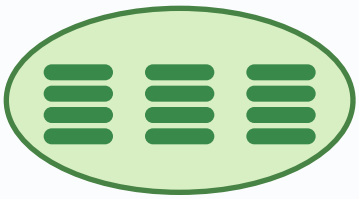


Water

+

Combine with

CHLOROPHYLL



Green pigment in chloroplasts
Simplified chloroplast diagram

Light-absorbing pigment

$C_6H_{12}O_6$



Atom inventory (not structure)

Glucose

O_2



Oxygen

LIGHT ENERGY



Sunlight

Atom colors: C = gray • O = red • H = blue

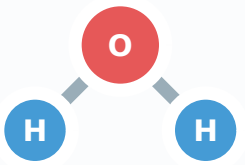
Student cutouts 2/3 • Print in color, single-sided, at 100%.

PHOTOSYNTHESIS • CUT & SORT

Cut on dashed lines. Keep all three sheets together.

Build a balanced equation. Place light and chlorophyll where they help.

H_2O



Water

→

Reaction arrow

O_2



Oxygen

O_2



Oxygen

CO_2



Carbon dioxide

CO_2



Carbon dioxide

Atom colors: C = gray • O = red • H = blue

Student cutouts 3/3 • Print in color, single-sided, at 100%.

TEACHER GUIDE

Photosynthesis: matter in, matter out, energy transferred

Copy: Print pages 1–3 for each group. Page 4 is for the teacher. Each complete set has 6 CO_2, 6 H_2O, 6 O_2, 1 glucose, 1 sun/light-energy card, 1 chlorophyll diagram, 2 plus signs, and 1 arrow.

Materials: Scissors, large chart paper, markers, and optional glue. Use sunlight or a lamp as a demonstration light source.

Student task: Sort the molecules into inputs and outputs. Use every molecule card to arrange a balanced photosynthesis equation. Place plus signs between molecule groups and the arrow between reactants and products. Place the sun/light-energy card and chlorophyll diagram above the arrow. Label the two sides “Reactants” and “Products.”

Answer key:
$6\ CO_2 + 6\ H_2O \rightarrow C_6H_{12}O_6 + 6\ O_2$
Light energy is absorbed by chlorophyll; show these above the arrow.

Check conservation: On each side, count 6 carbon atoms, 12 hydrogen atoms, and 18 oxygen atoms. The glucose picture is an atom inventory, not a structural model. Molecule pictures are simplified.

Short teacher script:
“Chlorophyll is a green pigment in chloroplasts that absorbs light energy.”
“Photosynthesis uses carbon dioxide and water to make glucose and oxygen.”
“Light provides energy. Chlorophyll helps absorb it; chlorophyll is not used up as a reactant.”
“Atoms are rearranged. Every atom on the reactant side must be accounted for on the product side.”
“Some of the absorbed light energy becomes chemical energy stored in glucose.”

Light-energy note: The single sun card represents the light energy needed for photosynthesis. Light energy is transferred and transformed; it is not an atom or a molecule. Do not include the sun or chlorophyll in the atom count.

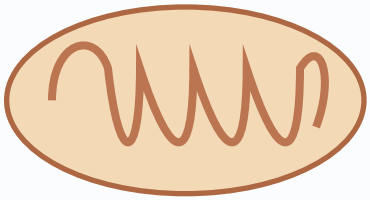
Quick discussion: Which cards represent matter? Which represent an energy source or an energy transfer? Why is chlorophyll placed above the arrow? How does the model show that matter is conserved?

CELLULAR RESPIRATION

Add-on cutouts • Use with your photosynthesis molecule cards

Cut on dashed lines. Each picture is a symbol, not a measured quantity.

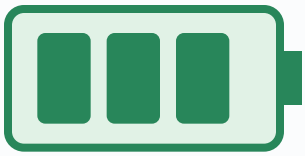
MITOCHONDRION



Most stages occur here

Simplified organelle diagram

ATP



Energy transferred to ATP

Usable energy for cell work

Battery = symbol for energy storage

HEAT



Energy released as heat

Transferred to the surroundings

Build the respiration model: Reuse all 19 molecule cards. Arrange glucose and 6 oxygen molecules on the reactants side; arrange 6 carbon dioxide and 6 water molecules on the products side.

Matter equation:
$C_6H_{12}O_6 + 6\ O_2 \rightarrow 6\ CO_2 + 6\ H_2O$

Add the symbols: Place the mitochondrion above the arrow. Place ATP and heat beside the products to show energy transfer. Set the sun and chlorophyll cards aside. The energy cards are annotations; do not include them in the atom count.

Teacher sentence: “Cellular respiration transfers some of the chemical energy in glucose to ATP, which cells use to do work, and releases some energy as heat.”

Accuracy note: Respiration begins in the cytoplasm; most later stages in plant and animal cells occur in mitochondria. ATP is made from ADP and phosphate, which this simplified model does not show. One ATP card does not mean that only one ATP molecule is produced.

Check: Both sides contain 6 carbon, 12 hydrogen, and 18 oxygen atoms. Plants carry out cellular respiration too.

TEACHER INSTRUCTIONS

Cellular respiration • Reuse the photosynthesis molecule cards

Build the respiration model: Reuse all 19 molecule cards. Arrange glucose and 6 oxygen molecules on the reactants side; arrange 6 carbon dioxide and 6 water molecules on the products side.

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